

MySmartStudy

RAGAS Evaluation Report

Quantitative Quality Assessment of the RAG Pipeline

An AI-Enhanced Collaborative Learning Management System

Final Year Project — UniKL MIIT

Generated 22 May 2026, 17:22

1. Overview

This report evaluates the Retrieval-Augmented Generation (RAG) pipeline of MySmartStudy using RAGAS (Retrieval-Augmented Generation Assessment), the standard open-source framework for measuring RAG quality. RAGAS scores each answer with an LLM acting as an impartial judge.

The pipeline was first evaluated as-shipped (baseline), then a set of targeted improvements was applied and the evaluation re-run. Both score sets are presented below so the impact of the tuning is transparent.

2. Methodology

Pipeline under test	app.rag_service.retrieve over per-course ChromaDB collections
Test samples	10
Judge model	gemini-2.5-flash
Embedding model	models/gemini-embedding-001
Evaluation time	504.2s

- Test questions were generated by an LLM from real indexed course material, with a reference answer grounded only in the source chunk.
- Each question was run through the actual backend RAG retrieval to fetch contexts, and an answer was generated from those contexts.
- RAGAS then scored every sample on five metrics using the Gemini judge model.

3. Improvements Applied

- Smaller, paragraph-aware chunks: chunk size 500 -> 220 tokens, overlap 50 -> 30, and a recursive splitter that respects paragraph and sentence boundaries instead of cutting mid-thought.
- Header-prefixed embeddings: each chunk is prefixed with its document title and type before embedding, so the semantic vector carries document-level context.
- Higher fetch ratio and top_k: over-fetch 6x (was 4x) before reranking, return top 8 (was 5) — gives the cross-encoder a wider candidate pool.
- Tightened answer prompt: the model is now instructed to answer ONLY what was asked, in 2-4 sentences, with no extra background or fabricated facts.

4. Aggregate Scores — Before & After

Metric	Before	After	Change	Meaning
Faithfulness	0.927	1.000	+0.073	How factually grounded the answer is in the retrieved context (no hallucination). Higher is better.

Answer Relevancy	0.689	0.744	+0.055	How directly the answer addresses the question. Higher is better.
Context Precision	0.787	0.970	+0.183	Whether the retrieved chunks are relevant and well-ranked. Higher is better.
Context Recall	0.900	0.900	+0.000	Whether retrieval captured all information needed to answer. Higher is better.
Answer Correctness	0.680	0.801	+0.121	Overall correctness of the answer vs the reference answer. Higher is better.

Overall mean: 0.797 -> 0.883 (+0.086) Strong — the RAG pipeline retrieves and answers reliably.

5. Per-Sample Results

#	Question	Faithfulness	Answer Relevan	Context Preci	Context Recall	Answer Correct
1	For the Tutorial Activity in Topic 2, identify the central debate topic and stat...	1.00	0.82	1.00	1.00	1.00
2	Based on the provided course material, identify the four key aspects that will b...	1.00	0.89	0.70	1.00	0.84
3	According to the text, what is fonologi the study of?	1.00	0.79	1.00	1.00	0.98
4	What bit length is stated for an IPv4 address in the provided material?	1.00	0.73	1.00	0.50	0.59
5	What is the derivative of $f(x) = x^3$?	1.00	0.74	1.00	1.00	0.54
6	What specific question did Muhammad Hafiz and Nurul Aisyah both ask?	1.00	0.86	1.00	1.00	0.89
7	How many steps are involved in an	1.00	0.81	1.00	1.00	0.61

	SN2 reaction?					
8	According to the text, what specific requirement does Tan Mei Ling inquire about...	1.00	0.89	1.00	1.00	0.84
9	Menurut teks ini, prinsip apakah yang digunakan oleh Stack?	1.00	0.90	1.00	0.50	0.99
10	Based on the provided quiz question and options, what is the keyword for definin...	1.00	0.00	1.00	1.00	0.74

6. Interpretation & Recommendations

Strongest metric: faithfulness (1.000). Weakest metric: answer relevancy (0.744).

Primary recommendation: Encourage the model to answer the question directly without padding.

Note: scores are computed by an LLM judge and carry some variance; a larger test set would tighten the estimates.

7. Conclusion

RAGAS provides an objective, repeatable measure of RAG quality for MySmartStudy. The scores above establish a quantitative baseline that can be tracked as the retrieval pipeline evolves.